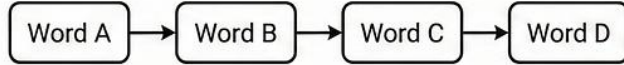


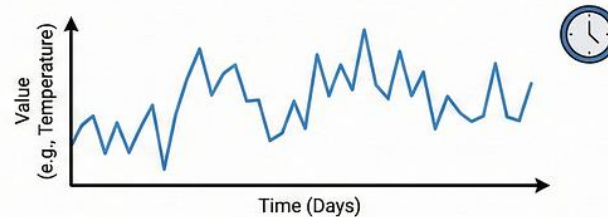
TYPES OF SEQUENCE / TIME-SERIES DATA

1. SEQUENCE DATA (NON-TEMPORAL)

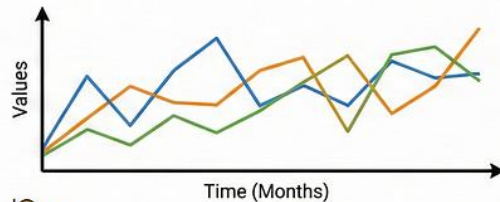


Examples: Natural Language, Genomic Sequences

2. TIME-SERIES DATA (UNIVARIATE)



3. TIME-SERIES DATA (MULTIVARIATE)



Examples: Sensor Readings, Stock Market Data

4. SPATIO-TEMPORAL DATA



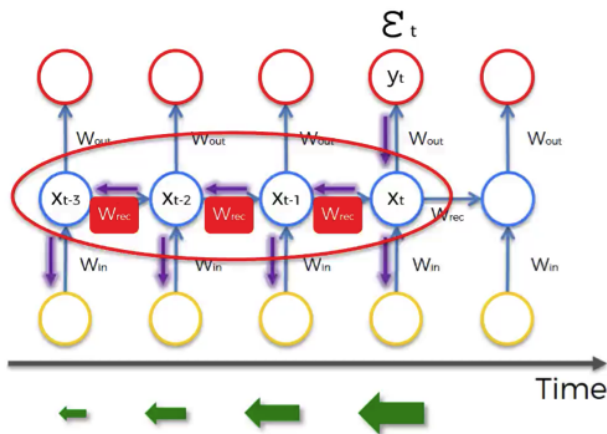
Examples: Traffic Flow, Weather Patterns, GPS Tracking

SEQUENCE DATA: WHY ORDER MATTERS

- Images are spatial; the relationship between pixels is based on their physical proximity.
- Time-series and sentences are sequential; the meaning of a word or data point depends entirely on what came before it.
- Standard Neural Networks (MLPs/CNNs) assume inputs are independent, but RNNs are designed to have "memory" of previous inputs.

VANISHING GRADIENTS: THE MEMORY LIMIT

The Vanishing Gradient Problem



$$\frac{\partial \mathcal{E}}{\partial \theta} = \sum_{1 \leq t \leq T} \frac{\partial \mathcal{E}_t}{\partial \theta} \quad (3)$$

$$\frac{\partial \mathcal{E}_t}{\partial \theta} = \sum_{1 \leq k \leq t} \left(\frac{\partial \mathcal{E}_t}{\partial \mathbf{x}_t} \frac{\partial \mathbf{x}_t}{\partial \mathbf{x}_k} \frac{\partial^+ \mathbf{x}_k}{\partial \theta} \right) \quad (4)$$

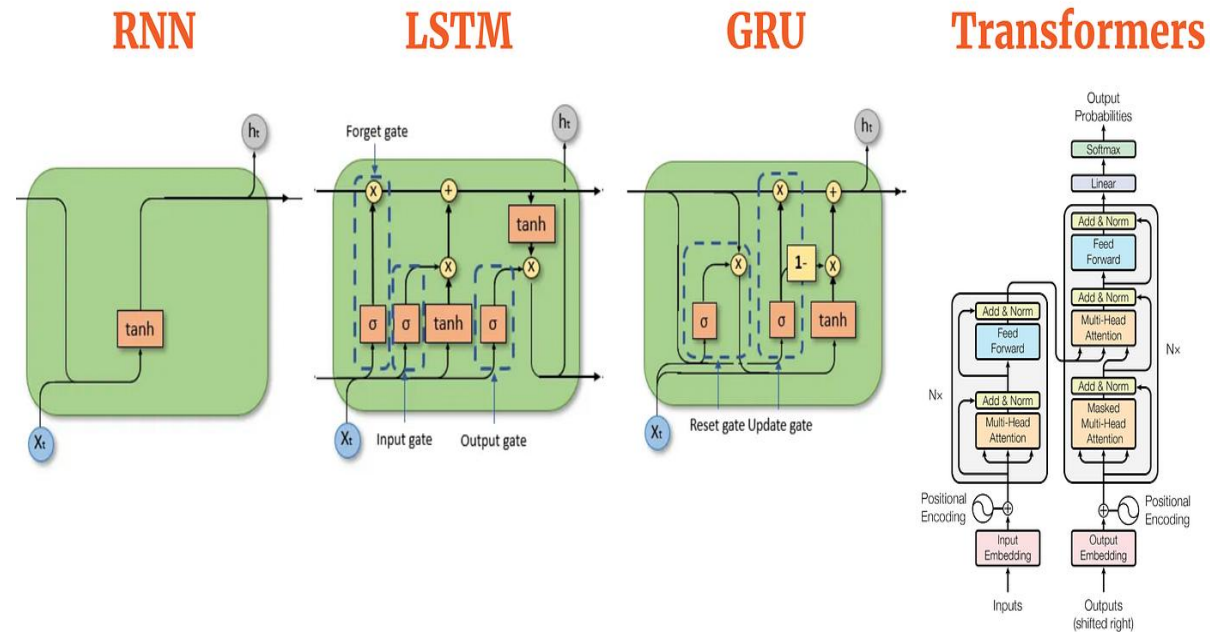
$$\frac{\partial \mathbf{x}_t}{\partial \mathbf{x}_k} = \prod_{t \geq i > k} \frac{\partial \mathbf{x}_i}{\partial \mathbf{x}_{i-1}} = \prod_{t \geq i > k} \mathbf{w}_{rec}^T \text{diag}(\sigma'(\mathbf{x}_{i-1})) \quad (5)$$

$W_{rec} \sim \text{small}$	$\Rightarrow$	Vanishing
$W_{rec} \sim \text{large}$	$\Rightarrow$	Exploding

Formula Source: Razvan Pascanu et al. (2013)

- RNNs process sequences by passing a "hidden state" from one step to the next.
- During training, gradients must be propagated back through every time step (Backpropagation Through Time).
- In long sequences, these gradients are multiplied repeatedly; if they are small, they "vanish" to zero.
- Result: Standard RNNs "forget" the beginning of a long sentence or pattern.

LSTMS & GRUS: GATING THE MEMORY



- **LSTM (Long Short-Term Memory):** Uses three "gates" (Forget, Input, Output) to regulate information flow.
- **Forget Gate:** Decides which part of the long-term memory is no longer relevant and should be erased.
- **GRU (Gated Recurrent Unit):** A simplified, faster version of LSTM that merges gates but maintains similar performance.
- These architectures allow the network to keep relevant information across hundreds of time steps.